Generative Project

Milestone 1 — Proposal

Flickr30k Text Embedding Baseline

Group 9 — Team and Roles

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GitHub repository: https://github.com/Sneha0504/IE7615_Generative_Project

Objective and Model

This project studies text conditioning and dataset adaptation for Stable Diffusion 1.5 (SD 1.5). The system has three pretrained components: a paired CLIP text encoder that turns a caption into a 77×768 token-embedding sequence, a latent-space U-Net that denoises the latent conditioned on those embeddings, and a VAE that decodes the final latent into a 512×512 image. Milestone 1 prepares a cleaned image–caption dataset, builds and verifies the frozen text→embedding pipeline, and runs five pretrained prompt→image sample tests. No weights are trained in this milestone.

Dataset and Preprocessing

We use a curated subset of **Flickr30k**: **3,000** unique image–caption pairs (seed 42) drawn from 17,500 candidate source captions. The notebook applies Unicode (NFKC) normalization, removes empty and duplicate captions (4 duplicates dropped), corrects EXIF orientation, de-duplicates images by pixel hash, and saves 512×512 RGB center crops, one caption per image. The custom image-disjoint split is **2400 / 300 / 300** train / validation / test pairs, with no image shared across splits.

Executed Environment and Sample Tests

The notebook ran on **CPU** (macOS, Python 3.12.2) with PyTorch 2.6.0, Diffusers 0.35.1, and Transformers 4.49.0. All **3,000** captions were encoded to float32 77×768 sequences (tokens 5–44, median 14; **0** truncated). **18/18 validation checks passed** — split isolation, finite and repeatable embeddings, empty and long prompts, tensor reload, five saved RGB 512×512 images, and correct prompt–seed mapping. The five prompt→image tests use 512×512 pixels, DDIM with 30 steps, CFG 7.5, and seeds 42–46; the safety checker is enabled and flagged none.

ID	Prompt	Seed	Seconds	Safety
p01	A brown dog jumps over a fallen log in a green park.	42	441	clear
p02	Two people in red jackets walk along a snowy mountain trail.	43	422	clear
p03	A cyclist in a yellow shirt rides past a brick building.	44	493	clear
p04	A child in a blue coat holds a red umbrella on a rainy street.	45	426	clear
p05	Three people play basketball on an outdoor court at sunset.	46	346	clear



Five prompt→image outputs (p01–p05), in prompt order without selection.

Observations and Next Steps

The outputs follow the prompts closely; p04 renders a blue umbrella instead of the requested red one, a visible attribute-binding error. Next, we will compare guidance scales and samplers on the validation prompts. These five outputs are illustrative only.